MSc Finance 20127

Corporate Finance (Business Valuation)
04 — From Cash Flows to Enterprise Value

WACC, APV, and the role of financing policy

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Where We Are in the Valuation Workflow

Equity story $\longrightarrow$ Financial statements $\longrightarrow$ Forecasts $\longrightarrow$ Cash flows $\longrightarrow$ **Discount rate** $\longrightarrow$
Terminal value $\longrightarrow$ Multiples $\longrightarrow$ Value bridge $\longrightarrow$ Recommendation

Today:
Projected FCFO $\longrightarrow$ Enterprise Value using WACC and APV

Today's Professional Task

Professional task:

- You have forecast FCFO
- Your job is to value the operating business and decide whether financing policy should be embedded in the discount rate or valued separately
  - match FCFO to the correct discount rate
  - estimate WACC under a target leverage policy
  - estimate APV when debt levels are known or changing
  - avoid double-counting tax shields and leverage risk
  - understand when WACC and APV should give similar or different answers

Plan for Today

Layer	What We Learn	Why It Matters
CORE	FCFO $\rightarrow$ Enterprise Value	All DCF valuation
CORE	WACC approach	Standard enterprise DCF
PRO	Choosing WACC vs APV	Project, M&A, LBO, leverage changes
TECH	β levering/unlevering, tax shields, bankruptcy costs	Avoid model inconsistency
PRACTICE	Discussion Questions	Turn formulas into valuation skill

From FCFO to Enterprise Value

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FCFO: From Earnings to Cash Flow (Reminder)

CORE

1. Free Cash flows to Equity (FCFE) — *Equity Side Valuation*
   - \+ Net Income — Adjusted
   - +/− Non Cash Expenditure (other than D&A) / Non Cash Revenues
   - − Net Capex (= Capital Expenditures − Depreciation/Amortization)
   - − Changes in non-cash Working Capital
   - − (Principal Debt Repayments − New Debt Issues)
   - = **Free Cash Flows to Equity**

2. Free Cash Flows from Operations (FCFO) — *Asset Side Valuation*
   - \+ FCFE
   - \+ Interest Expense (1 − Tax Rate)
   - \+ (Principal Debt Repayments − New Debt Issues)
   - = **Free Cash Flows from Operations (FCFO)**

   - \+ Operating Income (EBIT) — Adjusted
   - +/− Non Cash Expenditure (other than D&A) / Non Cash Revenues
   - − Taxes on EBIT
   - − Reinvestment Needs (Net Capital Expenditures and Working Capital)
   - = **Free Cash Flows from Operations (FCFO)**

[Ignoring preferred dividends. If preferred stock exist, preferred dividends will also need to be netted out]

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FCFO and Other Cash Flow Measures

PRO

Cash Flow Used	Definition	Use in Valuation
FCFO	Free Cash Flow from Operations	Discounting **FCFO** at **WACC** will yield the **value of operating assets** of a firm. Adding the value of non operating assets, if any, will yield firm value
FCFE	FCFO $-$ Interest $\times (1-\tau_C)$ $-$ Principal repaid $+$ Net debt issued $-$ Preferred dividends	Discounting **FCFE** at k_e will yield the **value of equity** of a firm
EBITDA	FCFO $+$ EBIT $\times \tau_C$ $+$ CAPEX $+ \Delta$ NWC	If you discount EBITDA at the cost of capital to value a firm, you are assuming that there are no taxes and that the firm will actively disinvest non-cash over time. It would be inconsistent to assume a growth rate or an infinite life for this firm
EBIT $\times (1-\tau_C)$ (*NOPAT is a modified version of this estimate and it removes non operating items that may affect reported EBIT*)	FCFO $+$ CAPEX $-$ D&A $+ \Delta$ non-cash NWC	If you discount the after-tax EBIT at the cost of capital to value a firm, you are assuming no reinvestment. The D&A is reinvested back into the firm to maintain existing assets. You can assume an infinite life but cannot assume growth

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FCFO: Cost of Capital and Adjusted Present Value

CORE

- Cost of Capital (CoC)
  - Firm value = **discount FCFO at** the Weighted Average Cost of Capital (**WACC**)

    Embedded in firm value are:
    (i) Tax benefits of debt (in the use of the after-tax cost of debt in the WACC)
    (ii) Expected additional risk associated with debt, i.e., higher k_e and k_d at higher D/E ratios

- Adjusted Present Value (APV)
  - Firm value = first **discount FCFO at k_u** (cost of unlevered equity, also called k_a). Then:
    (i) Add PV of interest tax savings generated by debt, and
    (ii) Subtract expected costs of bankruptcy

    Value includes separate estimates of:
    (i) Tax benefits of debt (PV of interest tax savings)
    (ii) Expected additional risk associated with debt (expected bankruptcy cost)

Depending on assumptions on **future growth**, can have 2-stage, 3-stage version
Same as with DDM and FCFE

FCFO: Cost of Capital and Adjusted Present Value

TECH

- The formula to unlever (and relever) beta depends on the assumed debt policy
- There are two main alternative approaches
  - Assumption 1: If debt as a % of total capital ($D/(D+E)$) is fixed (debt grows as the overall business grows):

$$\beta_{\text{levered}} = \beta_{\text{unlevered}} \times \left[1 + \frac{D}{E}\right]$$

  - Assumption 2: If debt is fixed in $ terms (as in Modigliani-Miller), we get Hamada formula:

$$\beta_{\text{levered}} = \beta_{\text{unlevered}} \times \left[1 + (1 - \tau_C) \cdot \frac{D}{E}\right]$$

Discussion Questions I

A privately held US firm operating in the food & beverage (F&B) business wants to estimate its cost of equity capital. To do so, it collected the following data:

Inputs	Values	Inputs	Values
Interest Rate on $-denominated T-Bonds	3%	Shareholders' Equity	$100m
Equity Risk Premium (ERP) on US market	6%	Net Debt	$70m
Unlevered Beta for Food & Beverage sector	0.5	Tax Rate	20%

The CFO aims at a constant debt policy in monetary terms (i.e., fixed debt)

i. What is the cost of equity of this firm?

Now the CFO aims at a constant leverage financing policy

ii. What is the cost of equity of this firm?

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WACC: Financing Policy Inside the Discount Rate

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The Cost of Capital Approach: Stable Growth Model

CORE

- A firm with FCFO growing at a stable growth rate (steady-state) can be valued as:

$$\text{Firm Value}_0 = \frac{\text{FCFO}_1}{\text{WACC} - g_n}$$

where:

- FCFO_1 = Expected FCFO next year
- WACC = Weighted Average Cost of Capital
- g_n = Annual Growth rate of FCFO forever. Under a stable growth rate assumption:
  - g_n = Reinvestment rate $\times$ Returns on Capital
  - The stable growth rate used in the model has to be $\leq$ the growth rate in the economy (nominal or real depending on CF and WACC)
  - Firm characteristics have to be consistent with assumptions of stable growth

- Limitations
  - Like all stable growth models, it is sensitive to assumptions about g_n
  - Such sensitivity is higher when using WACC (which is lower than k_e)
  - It is sensitive also to assumptions made about CAPEX relative to depreciation
  - Cost of capital should also reflect a stable growth firm (β, D/E)

Discussion Questions II

Telesp provides telecommunication services to Sao Paulo. Its financial data are :

Inputs	Values	Inputs	Values
EBIT	R$3,544m	BV Equity (year -1)	R$20,057m
Effective tax rate	30.00%	BV Debt (year -1)	R$8,042m
CAPEX	R$1,659m	Cash (year -1)	R$4,277m
D&A	R$1,914m	Market Value Equity	R$21,982m
Δ non-cash NWC	R$1,119m	Market Value Net Debt	R$5,519m

iii. What is the expected growth rate of Telesp?

The risk free rate in Brazil is 7%, the ERP is 8%, the industry β is 0.8, the pre-tax k_d is 9.50%

iv. What is the WACC for Telesp?

v. What is the enterprise value and the equity value of Telesp?

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The Cost of Capital Approach: General Model

CORE

- The value of the firm, in the most general case, is the present value of expected FCFOs:

$$\text{Firm Value}_0 = \sum_{t=1}^{\infty} \frac{\mathbb{E}\left[\text{FCFO}_t\right]}{(1+\text{WACC})^t}$$

where:

$\mathbb{E}\left[\text{FCFO}_t\right]$: Expected Free Cash Flow from Operations in period t
WACC: Weighted Average Cost of Capital

- If the firm reaches steady state after n years and then grows at a stable g_n after that:

$$\text{Firm Value}_0 = \sum_{t=1}^{n} \frac{\mathbb{E}\left[\text{FCFO}_t\right]}{\left(1+\text{WACC}_{hg}\right)^t} + \frac{\frac{\mathbb{E}[\text{FCFO}_{n+1}]}{\text{WACC}_{st}-g_n}}{\left(1+\text{WACC}_{hg}\right)^n}$$

where:

$\mathbb{E}\left[\text{FCFO}_t\right]$: Expected Free Cash Flow from Operations in period t
WACC: Weighted Average Cost of Capital (*hg* high growth; *st* stable)
g_n: Stable growth rate

The Cost of Capital Approach: General Model

PRO

- Stable growth rate
  - g_n = Reinvestment Rate $\times$ Return on Capital
  - The stable growth rate has to be reasonable relative to the nominal growth rate in the economy/industry in which the firm operates
  - The stable growth rate should be linked to fundamentals
- Limitations
  - FCFEs are a much more intuitive measure of cash flows than FCFO. Furthermore, FCFE is a real cash flow that can be traced and analyzed, whereas FCFO is a theoretical construct
  - Its focus on pre-debt cash flows might hide survival problems if firms cannot raise additional equity to pay down debt
  - Using a debt ratio in the cost of capital to incorporate the effect of leverage requires to assume firms can freely issue debt and equity and rebalance their capital structure, which might not always be reasonable
- Model works best for firms
  - Reaching the steady state in several stages (two, three, n)
  - Having either very high or very low leverage, or in the process of changing their leverage. In such cases, calculating FCFEs is much more challenging because of the volatility induced by debt (net) payments

Using FCFO in CoC Approach: Comparison with FCFE — I

PRO

- The CoC model with FCFO, unlike the model with FCFE or the DDM (coming up), values the whole firm rather than its sole equity
  - To extract the value of equity need to subtract from firm value the market value of outstanding (net) debt
  - Equity value under firm valuation (FCFO) and equity valuation (FCFE) will be the same if we make consistent assumptions about financial leverage (in practice, this is hard)

- Relative to FCFE, using FCFO and thus valuing the whole firm has the advantage that CF relating to debt do not have to be considered explicitly, since the FCFO is a pre debt CF; whereas, they have to be taken into account in estimating FCFE

- On the other hand, the firm valuation approach does require information about
  (i) Debt ratios, and
  (ii) Interest rates to estimate the WACC

Discussion Questions III

Consider a no-growth, no reinvestment perpetual firm with these inputs:

Inputs	Values
EBIT	$166.67m
Effective tax rate	40.00%
Equity (*market value*)	$600m
k_e	13.87%
Debt (*market value*)	$400m
k_d (*pre-tax*)	7%

vi. What is the WACC of this firm?

vii. What is the enterprise value of this firm?

viii. What is the equity value if we subtract debt from the enterprise value?

ix. What is the equity value if we discount net income at k_e?

Using FCFO in CoC Approach: Comparison with FCFE — II

PRO

- The previous case works because of three assumptions:
  1. Values for debt and equity
     Debt and equity values used to compute the WACC were the same values obtained in the valuation. This circular reasoning (you need WACC to obtain the values in the first place) indicates that a WACC based on market value weights will not yield the same equity value as an equity valuation model, if the firm is not fairly priced to begin with
  2. Extraordinary items
     There are no extraordinary or non-operating items that affect net income but not operating income. Thus net income equals operating income minus interest expenses and taxes
  3. Interest expenses
     Interest expenses equal pre-tax cost of debt times market value of debt. If a firm has old debt on its books, with interest expenses differing from the current pre-tax cost of debt, the two approaches will diverge
- If there is expected growth, the potential for inconsistency increases
  - Need to ensure borrowing enough money to fund new investments to keep debt ratio at a level consistent with what that used in computing WACC

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Discussion Questions IV

DaimlerChrysler is a mature firm in a mature industry

$\Longrightarrow$ it is in stable growth

$\Longrightarrow$ it aims to keep a constant leverage ratio

Inputs (figures in DM miln)	Values
EBIT (operating income)	9,324
Effective tax rate	46.94%
ROC (after tax)	7.15%
Equity (market value)	62,300
Debt (market value)	64,500
Cash & marketable securities	18,068
Bottom up unlevered β for industry	0.61
Daimler rating	AAA
Long-term German Bond rate (in DM)	4.87%
Default spread	0.20%
ERP	4.00%
Long-term growth rate	3.00%

x. What is the reinvestment rate of DaimlerChrysler?

xi. What are the cost of equity, cost of debt, and WACC of DaimlerChrysler?

xii. What are the enterprise value and the equity value of DaimlerChrysler?

APV: Financing Policy Valued Separately

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APV in One Line

CORE

$$\mathbf{V}_l = \mathbf{V}_u + \mathbf{TB} - \mathbf{BC}$$

$\mathbf{V}_l$: Value of Levered Firm

$\mathbf{V}_u$: Value of Unlevered Firm

TB: Present Value of Tax Benefits of Debt, aka Tax Shields

BC: Present Value of Bankruptcy Costs

APV: General Model — I

PRO

The APV estimates the **Value of Levered Firm, $V_l = V_u + TB - BC$**, where:

1. **Value of unlevered firm, V_u**

$$V_{u,0} = \frac{\mathbb{E}\left[FCFO_1\right]}{(\rho_u - g_n)} \qquad \text{or} \qquad V_{u,0} = \sum_{t=1}^{n} \frac{\mathbb{E}\left[FCFO_t\right]}{(1+\rho_u)^t} + \frac{\frac{\mathbb{E}[FCFO_{n+1}]}{\rho_{u,st} - g_n}}{(1+\rho_{hg})^n}$$

where:

$\mathbb{E}\left[FCFO_t\right]$: Expected Free Cash Flow from Operations in period t

ρ_u: Unlevered k_e (*hg* high growth; *st* stable)

g_n: Stable growth rate

2. **Expected tax benefit from borrowing, TB**. Two cases:

i. D is low and fixed: $TB_0 = (\tau_C \cdot k_d \cdot D) / k_d = \tau_C \cdot D$

ii. D is high and/or D/E is predetermined: $TB_0 = \sum_{t=1}^{n} \frac{\mathbb{E}[\tau_C \cdot k_d \cdot D]}{(1+\rho_u)^t} + \frac{\tau_C \cdot k_d \cdot D}{(\rho_u - g_n)\cdot(1+\rho_u)^n}$

3. **Expected bankruptcy costs, BC**

$$BC_0 = \pi_d \times \text{LGD}$$

where:

π_d: Probability of default

LGD: Loss Given Default

APV: General Model — II

PRO

There are two ways in which π_d (probability of default) can be indirectly estimated:

1. Estimate a bond rating; use the empirical estimates of default probabilities for the rating
2. Use a statistical approach, e.g., a probit, to estimate the probabilities of default, based on the firm's observable characteristics, at each debt level

- The bankruptcy cost can be estimated, albeit with considerable error, from studies that have looked at the magnitude of this cost in actual bankruptcies:
  - Research looking at direct bankruptcy costs concludes these are small relative to firm value
  - *Indirect* costs of bankruptcy can be substantial, but they vary widely across firms. Shapiro (1989) and Titman (1984) conclude that the indirect costs could be as large as 25%-30% of firm value but provide no direct evidence of this

APV: General Model — III

PRO

Probability of default over 10 years by bond rating class, from Altman (2009)

Rating	Probability of Default
AAA	0.07%
AA	0.51%
A+	0.60%
A	0.66%
A-	2.50%
BBB	7.54%
BB+	10.00%
BB	16.63%
B+	25.00%
B	36.80%
B-	45.00%
CCC	59.01%
CC	70.00%
C	85.00%
D	100.00%

APV: General Model — IV

TECH

- Stable growth rate
  - g_n = Reinvestment Rate $\times$ Return on Capital
  - The stable growth rate has to be reasonable relative to the nominal growth rate in the economy/industry in which the firm operates
  - The stable growth rate should be linked to fundamentals
- Unlevered Cost of Equity, ρ_u
  - In the first step, to estimate the unlevered k_e, we draw on our earlier analysis and compute the firm's unlevered beta:

$$\beta_{\text{unlevered}} = \beta_{\text{levered}} / \left[1 + \frac{D}{E}\right]$$

or

$$\beta_{\text{unlevered}} = \beta_{\text{levered}} / \left[1 + (1 - \tau_C) \cdot \frac{D}{E}\right]$$

depending on assumptions about financial policy (see above). Then use unlevered β to compute cost of equity, e.g.,

$$\rho_u = rf + \beta_u \cdot \text{ERP}$$

- Limitations
  - In the second step, the tax rate used should be the firm's marginal tax rate and it is assumed to stay constant over time. If you anticipate the tax rate changing over time, you can still compute the PV of tax benefits over time, but you cannot use the previous perpetual growth equation;
  - Need to modify this equation to estimate TB if the current interest expenses do not reflect the current cost of debt;
  - Estimating π_d and BC is challenging and subject to errors

APV: Advantages and Challenges — I

PRO

- Advantage of APV:
  - (i) It separates the effects of debt into its different components
  - (ii) It allows the analyst to use different discount rates for each component
  - (iii) We don't need to assume that the debt ratio stays constant forever, unlike CoC
  - (iv) We can compute costs and benefits of debt by keeping fixed the dollar value of debt

- Challenges of APV:
  - (i) It is hard to estimate the probabilities of default and cost of bankruptcy
  - (ii) As a result, many analyses ignore bankruptcy costs $\Longrightarrow$ firm value always increases as firms borrow more. In turn $\Longrightarrow$ firms' optimal debt ratio should be 100% debt (is this realistic?)

- In general, under the same assumptions, APV and CoC give very similar answers. However, the APV approach is more practical when firms are evaluating a "stable" dollar amount of debt, whereas the CoC approach is more appropriate when firms keep stable debt proportions D/E

APV: Advantages and Challenges — II

PRO

CoC and APV may sometimes yield different values, for these reasons:

1. Bankruptcy costs

   The two models consider bankruptcy costs very differently: the APV provides more flexibility in allowing consideration of indirect bankruptcy costs. To the extent that these costs do not show up or show up inadequately in the pretax cost of debt, the APV approach will yield a more conservative estimate of value

2. Tax benefit

   The APV considers the tax benefit from a dollar debt value, usually based on existing debt. The CoC estimates the tax benefit from a debt ratio that may require the firm to borrow increasing amounts in the future

3. Financial leverage

   The COC approach is a more practical choice when valuing ongoing firms that are not going through sharp changes on financial leverage; it is easier to work with a debt ratio than with dollar-debt levels. The APV approach is more useful for transactions funded disproportionately with debt and where debt repayments are negotiated or known

4. Discount rate

   The APV approach often uses the pre-tax cost of debt as the discount rate to estimate the value of the tax savings from debt; there are variations on the APV that discount the tax savings back at the WACC or the unlevered k_e that yield values that are closer to those obtained in the CoC approach

Discussion Questions V

J. Crew is a U.S. retailer that sells clothes made under its brand name through its own stores and online. It was acquired in a leveraged deal for $2.7 bn, with $1.85 bn coming from debt (rated BB) and a pre-tax k_d of 7%. Its β_U is 1, r_f is 3.5%, the ERP is 5%, tax rate is 35%, and g is 3.5%. J. Crew generated $230 mln in EBIT on revenues of $1,722 mln before acquisition. with a ROC of 14%. Total bankruptcy costs are 30% of firm value and the acquisition increased the probability of bankruptcy by 20%. Below the debt schedule of J. Crew in $ mln

Year	Debt Due
1	$1,850
2	$1,700
3	$1,550
4	$1,400
5	$1,250
6	$1,100
7	$950
8	$800
9	$650
10	$500
Perpetuity	$500

xiii. What is the unlevered cost of equity, k_U, of J. Crew?

xiv. What is the unlevered value of J. Crew?

xv. What are the expected bankruptcy costs?

xvi. What is the levered value of J. Crew? How does it compare with the price of $2.7 bn?

WACC or APV? A Practical Choice

Situation	Better Tool	Why
Stable target D/E or D/(D+E)	WACC	Financing policy embedded in WACC
Known/predetermined debt schedule	APV	Tax shields can be valued directly
LBO/Acquisition financing	APV	Debt repayment path is explicit
Changing leverage	APV or staged WACC	Single constant WACC misleading
Mature firm, stable leverage	WACC	Simpler and standard
Financial distress/high leverage	APV	Bankruptcy costs modeled explicitly

Common Mistakes in WACC and APV Valuation

1. Discounting FCFO at the cost of equity
2. Discounting FCFE at WACC
3. Using book-value capital weights without justification
4. Using today's leverage in WACC when the forecast assumes a different target leverage
5. Applying a stable-growth terminal value while β, leverage, ROIC, or reinvestment assumptions are not stable
6. Double-counting tax shields: once in WACC and again separately
7. Treating WACC and APV as different answers rather than different implementations of financing assumptions
8. Using APV but ignoring bankruptcy costs, mechanically making more debt always better

Takeaways

1. FCFO values the operating business; financing effects must be handled consistently
2. WACC embeds financing effects in the discount rate
3. APV values the unlevered business first and then adds tax shields and subtracts expected distress costs
4. WACC works best with stable target leverage ratios
5. APV works best with known or changing dollar debt levels, especially leveraged transactions
6. WACC and APV should give similar answers only under consistent assumptions
7. The main valuation errors come from mixing cash flows, discount rates, and financing-policy assumptions